Wenhan Cao

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Research Interests

My research develops a **probabilistic foundation for physical AI**: embodied systems, such as robots and autonomous vehicles, that must perceive, decide, and act under uncertainty. I design learning, estimation, and control methods that turn uncertainty from a nuisance into a first-class computational object. My long-term goal is to make probability the common language of reliable autonomy—enabling physical AI to reason about what it knows, adapt to what it doesn't, and earn human trust.

Work Experience

National University of Singapore — August 2025 – Present
Eric and Wendy Schmidt AI in Science Postdoctoral Fellow,
Department of Electrical and Computer Engineering — *Singapore*
Supervisor: Prof. Lin Zhao

Education

Tsinghua University — September 2019 – June 2025
Ph.D., School of Vehicle and Mobility — *Beijing, China*
Supervisors: Prof. Shengbo Eben Li & Prof. Chang Liu

University of Manchester — January 2023 – July 2024
Visiting Ph.D. Researcher in Computer Science — *Manchester, UK*
Supervisor: Prof. Wei Pan

Technical University of Munich — September 2023 – December 2023
Research Visit in Optimization and Control — *Munich, Germany*
Supervisor: Prof. Sandra Hirche

Beijing Jiaotong University — September 2015 – June 2019
B.Eng. in Electrical Engineering — *Beijing, China*
GPA ranking: 1/305

Selected Publications

* *denotes equal contribution. A full list is available in the publication list or on Google Scholar.*

[1] Shiqi Liu*, **Wenhan Cao***, Chang Liu, Zeyu He, Tianyi Zhang, Yinuo Wang & Shengbo Eben Li. *One Filters All: A Generalist Filter for State Estimation*. In Advances in Neural Information Processing Systems (**NeurIPS**), 2025. [Paper] [Poster]

[2] **Wenhan Cao**, Alexandre Capone, Rishabh Dev Yadav, Sandra Hirche & Wei Pan. *Computation-Aware Learning for Stable Control with Gaussian Process*. In Robotics: Science and Systems (**RSS**), 2024. [Paper] [Poster] [Recording]

[3] **Wenhan Cao** & Wei Pan. *Impact of Computation in Integral Reinforcement Learning for Continuous-Time Control*. In International Conference on Learning Representations (**ICLR**), 2024. **(Spotlight)**. [Paper] [Poster] [Code]

[4] **Wenhan Cao**, Tianyi Zhang, Zeju Sun, Chang Liu, Stephen S.-T. Yau & Shengbo Eben Li. *Nonlinear Bayesian Filtering with Natural Gradient Gaussian Approximation*. IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2026. [Paper] [Code] [Slides]

[5] **Wenhan Cao**, Shiqi Liu, Chang Liu, Zeyu He, Stephen S.-T. Yau & Shengbo Eben Li. *Convolutional Bayesian Filtering*. IEEE Transactions on Automatic Control (**TAC**), 2026. [Paper] [Slides]

[6] **Wenhan Cao**, Chang Liu, Zhiqian Lan, Shengbo Eben Li, Wei Pan & Angelo Alessandri. *Robust Bayesian Inference for Moving Horizon Estimation*. **Automatica**, 173, 112108, 2025. [Paper] [Code]

[7] Jingliang Duan, **Wenhan Cao**, Yang Zheng & Lin Zhao. *On the Optimization Landscape of Dynamic Output Feedback Linear Quadratic Control*. IEEE Transactions on Automatic Control (**TAC**), 69(2):920–935, 2024. [Paper] [Code]

[8] Kangjie Zhou, Zhejia Wen, Zhiyong Zhuo, Zike Yan, Pengying Wu, Shuaiyang Li, Han Gao, Kang Ding, **Wenhan Cao**, Wei Pan & Chang Liu. *CoINS: Counterfactual Interactive Navigation via Skill-Aware VLM*. IEEE Transactions on Automation Science and Engineering (**T-ASE**), 2026. [Paper] [Project]

[9] Tianyi Zhang, **Wenhan Cao**, Chang Liu, Tao Zhang, Jiangtao Li & Shengbo Eben Li. *Robust State Estimation for Legged Robots with Dual Beta Kalman Filter*. IEEE Robotics and Automation Letters (**RA-L**), 10(8):7955–7962, 2025. [Paper]

[10] Shiqi Liu, **Wenhan Cao**, Chang Liu, Tianyi Zhang & Shengbo Eben Li. *Convolutional Unscented Kalman Filter for Multi-Object Tracking with Outliers*. IEEE Transactions on Intelligent Vehicles (**T-IV**), 2024. [Paper]

[11] Tianyi Zhang, **Wenhan Cao**, Chang Liu, Feihong Zhang, Wei Wu & Shengbo Eben Li. *NANO-SLAM: Natural Gradient Gaussian Approximation for Vehicle SLAM*. In IEEE International Conference on Intelligent Transportation Systems (**ITSC**), 2025. [Paper]

[12] Tianyi Zhang, **Wenhan Cao** & Shengbo Eben Li. *Natural Gradient Gaussian Approximation Filter with Positive Definiteness Guarantee*. In American Control Conference (**ACC**), 2026.

[13] **Wenhan Cao**, Chang Liu, Zhiqian Lan, Yingxi Piao & Shengbo Eben Li. *Generalized Moving Horizon Estimation for Nonlinear Systems with Robustness to Measurement Outliers*. In American Control Conference (**ACC**), 2023. [Paper] [Code] [Slides]

[14] **Wenhan Cao**, Jingliang Duan, Shengbo Eben Li, Chang Liu, Chen Chen & Yu Wang. *Primal-Dual Estimator Learning Method with Feasibility and Near-Optimality Guarantees*. In IEEE Conference on Decision and Control (**CDC**), 2022. [Paper] [Slides]

[15] **Wenhan Cao**, Jianyu Chen, Jingliang Duan, Shengbo Eben Li & Yao Lyu. *Reinforced Optimal Estimator*. In Modeling, Estimation and Control Conference (**MECC**), 2021. **(Student Best Paper Finalist)**. [Paper] [Slides]

PREPRINTS

[P1] Keyu Yan, Xuyang Chen, **Wenhan Cao** & Lin Zhao. *Discriminating Out-of-Distribution Actions in Offline Reinforcement Learning via Quantile Advantage*. Under review at NeurIPS 2026. [Code]

[P2] **Wenhan Cao**, Keyu Yan, Xuyang Chen & Lin Zhao. *Trajectory Editing as Posterior Inference for Offline Reinforcement Learning*. Under review at NeurIPS 2026. [Code]

[P3] Xuyang Chen, Keyu Yan, Danze Chen, Xiang Zheng, **Wenhan Cao** & Lin Zhao. *Learning Robotic Skills from Offline Preference-based RL*. Under review at NeurIPS 2026. [Code]

[P4] **Wenhan Cao**, Xuyang Chen, Shuyuan Wang & Lin Zhao. *How Much Information is Needed for Accurate Kalman Filtering?* Under review at NeurIPS 2026. [Code]

[P5] Rui Huang, Xin Chen, **Wenhan Cao**, Lidong Li, Yizhe Xiao, Yichao Gao, Yuqi Shi & Lin Zhao. *Agile Modular Aerial Robot Load Transport with Arbitrary Configurations*. Under review at RSS 2026.

[P6] Han Gao, **Wenhan Cao**, Ieng Hou U, Kangjie Zhou, Ying Sun, Angelo Alessandri & Chang Liu. *Chance-Constrained Gaussian Filtering*. Under review at Automatica.

Honors & Awards

Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship 2026
Study Abroad Fund, Tsinghua University 2022
Student Best Paper Finalist, Modeling, Estimation and Control Conference 2021
China National Scholarship 2016
First Prize Scholarship, Beijing Jiaotong University 2016, 2017 & 2018

Software

General Optimal control Problems Solver (GOPS) — an easy-to-use reinforcement learning (RL) solver package designed to build real-time, high-performance controllers for industrial applications. I was primarily responsible for the core design and implementation of the **trainer**, **sampler**, and **buffer** modules. [Docs] [Code] [Paper]

Selected Research Projects

RL Post-Training of MindGPT November 2024
Project Member, Internship at Li Auto

Reinforcement Learning for Autonomous Driving Decision and Control on Open Roads May 2024
Project Member — Supported by Toyota & IdriverPlus.

State Estimation for Warehouse Automated Logistics Vehicles May 2022
Student Project Leader — Supported by Geek+; the technology has been implemented in the company's product lineup.

Networked Modeling and Cooperative Control of Connected and Automated Electric Vehicles November 2020
Student Project Leader — Supported by MOST.

Invited Talks

NANO Filter: Bayesian Filtering with Natural Gradient Gaussian Approximation. Department of Astronomy, Tsinghua University, Beijing, China, hosted by Prof. Zheng Cai, August 2024.

Convolutional Bayesian Filtering. Department of Mathematical Sciences, Tsinghua University, Beijing, China, hosted by Prof. Stephen S.-T. Yau, February 2024.

Learning-based State Estimation Methods. Technical University of Munich, Munich, Germany (online), hosted by Prof. Sandra Hirche, February 2023.

Teaching Experience

Tsinghua University, School of Vehicle and Mobility February 2022 – May 2022
Teaching Assistant, 150051 Intelligent Vehicles

Professional Services

Conference Reviewer: NeurIPS, ICML, ICLR, AAMAS, L4DC, CDC, ACC & IFAC NMPC
Journal Reviewer: TPAMI, T-RO, TAC, Automatica, RA-L, T-ITS, T-ASE, TII & TNNLS